Quantile GAM Regression

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Introduction

Standard GAMs model the **conditional mean** $E[Y \mid X]$. **Quantile regression** instead models conditional quantiles $Q_\tau(Y \mid X)$, providing a more complete picture of the response distribution—especially useful when:

- The variance changes with covariates (heteroscedasticity)
- Interest lies in tail behavior (e.g., 90th percentile)
- The distribution is skewed

GAM.jl implements quantile GAMs following the **qgam** approach (Fasiolo et al., 2021), using the **Extended Log-F (ELF)** loss as a smooth approximation to the pinball (check) loss:

$$\rho_\tau(u) = u(\tau - \mathbb{1}_{u<0})$$

The ELF family provides a smooth, twice-differentiable loss that converges to the pinball loss, enabling standard penalized likelihood fitting with automatic smoothing parameter selection.

Setup

```
using GAM
using StatsAPI: predict, fitted, residuals
using DataFrames
using CSV
using Random
using Statistics
using Plots
```

Simulate heteroscedastic data

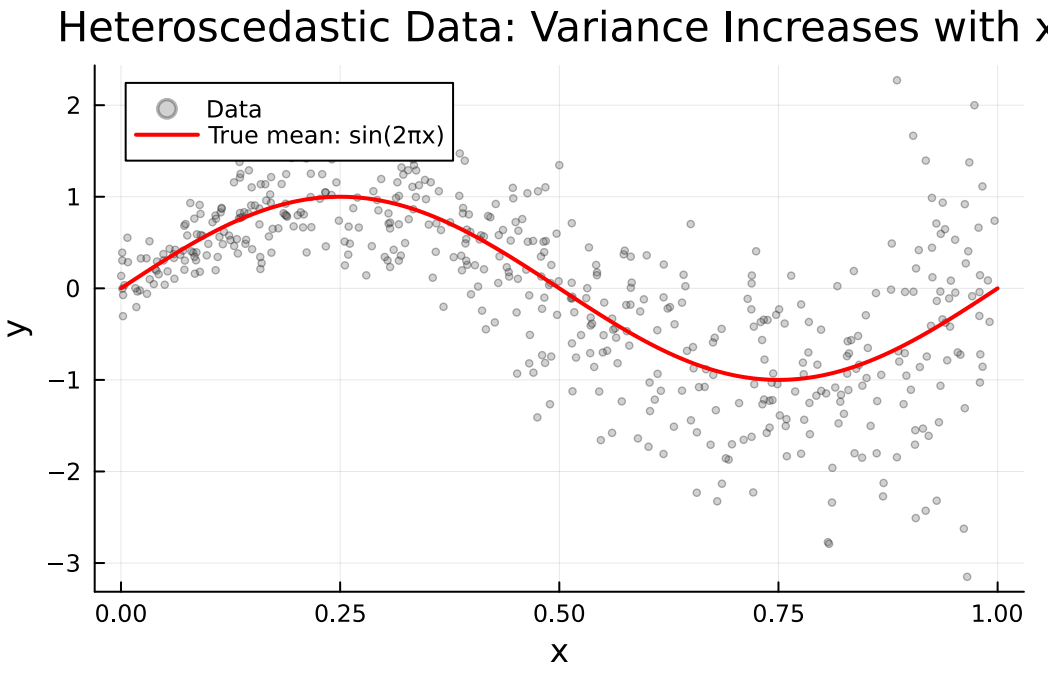
We create data where the conditional variance increases with x:

$$y = \sin(2\pi x) + (0.2 + 0.8x)\varepsilon, \quad \varepsilon \sim N(0, 1)$$

```
df = CSV.read("data.csv", DataFrame)
x = df.x; y = df.y
n = nrow(df)
println("Data: n = $n")
println("y range: [$(round(minimum(y), digits=2)), $(round(maximum(y), digits=2))]")
println("Mean y: $(round(mean(y), digits=3))")
```

```
Data: n = 500
y range: [-3.15, 2.27]
Mean y: -0.002
```

```
scatter(x, y, color=:grey40, markersize=2, alpha=0.3,
    xlabel="x", ylabel="y",
    title="Heteroscedastic Data: Variance Increases with x",
    label="Data", legend=:topleft)
x_grid = range(0, 1, length=200)
plot!(x_grid, sin.(2 .* x_grid), color=:red, linewidth=2, label="True mean: sin(2 x)")
```



Fit a single quantile GAM

The `qgam` function fits a GAM for a single quantile τ. Internally it:

1. Fits a preliminary Gaussian GAM to estimate variance
2. Computes the smoothness constant
3. Tunes the learning rate via grid search
4. Fits the final model with the ELF family

Median regression (= 0.5)

```
m_50 = qgam(@gam_formula(y ~ s(x, k=20, bs=:cr)), df, 0.5)
```

```
Generalized Additive Model

Formula: y ~ 1

Family: ELF
Link:   IdentityLink
Method: REML
```

```
Parametric coefficients:

                   Coef.  Std. Error      z  Pr(>|z|)

(Intercept)  -0.00166205    0.135095  -0.01    0.9902


Approximate significance of smooth terms:

Smooth                     edf   Ref.df

s(x,bs=cr)                4.11       19


R² (adj) = 0.554   Deviance explained = 53.9%
n = 500
```

```
yhat_50 = predict(m_50)
println("Median regression fitted range: [$(round(minimum(yhat_50), digits=3)), $(round(maxi
```

```
Median regression fitted range: [-1.072, 0.967]
```

Compare with mean regression

```
m_mean = gam(@gam_formula(y ~ s(x, k=20, bs=:cr)), df)
yhat_mean = predict(m_mean)

println("Mean regression fitted range: [$(round(minimum(yhat_mean), digits=3)), $(round(maxi
println("Correlation (mean vs median): $(round(cor(yhat_mean, yhat_50), digits=4))")
```

```
Mean regression fitted range: [-1.069, 0.968]
Correlation (mean vs median): 1.0
```

For symmetric errors, mean and median regression give similar results. With heteroscedastic or skewed data, they can diverge.

Fit multiple quantiles

Individual quantile fits

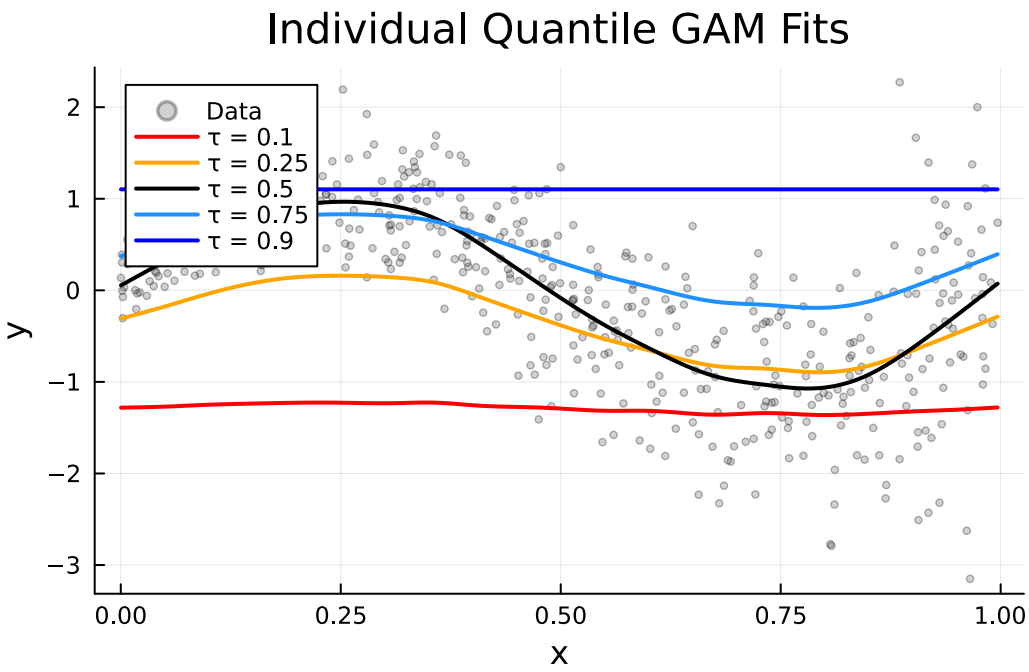
```
quantiles = [0.1, 0.25, 0.5, 0.75, 0.9]

fits = Dict{Float64, Any}()
for qu in quantiles
    fits[qu] = qgam(@gam_formula(y ~ s(x, k=20, bs=:cr)), df, qu)
end
```

Examine fitted ranges for each quantile:

```
ord = sortperm(x)
for qu in quantiles
    yhat = predict(fits[qu])
    println("  = $qu: fitted range [$(round(minimum(yhat), digits=3)), $(round(maximum(yhat),
end
```

```
  = 0.1: fitted range [-1.362, -1.225]
  = 0.25: fitted range [-0.895, 0.159]
  = 0.5: fitted range [-1.072, 0.967]
  = 0.75: fitted range [-0.191, 0.831]
  = 0.9: fitted range [1.103, 1.103]
```

```
qcolors = Dict(0.1 => :red, 0.25 => :orange, 0.5 => :black, 0.75 => :dodgerblue, 0.9 => :blue
scatter(x, y, color=:grey40, markersize=2, alpha=0.3,
    xlabel="x", ylabel="y", title="Individual Quantile GAM Fits",
    label="Data", legend=:topleft)
for qu in quantiles
    yhat = predict(fits[qu])[ord]
    plot!(x[ord], yhat, color=qcolors[qu], linewidth=2, label="  = $qu")
end
current()
```



Using `mqgam` for simultaneous fitting

`mqgam` fits all quantiles simultaneously, sharing the preliminary variance estimate for efficiency:

```
mq = mqgam(@gam_formula(y ~ s(x, k=20, bs=:cr)), df, quantiles)
```

```
(fits = Dict{Float64, Any}(0.5 => GamModel(n_smooth=1, edf=5.1, deviance=8.75), 0.9 => GamMo
```

```
println("Quantiles fitted: ", mq.quantiles)
println("Number of models: ", length(mq.fits))
```

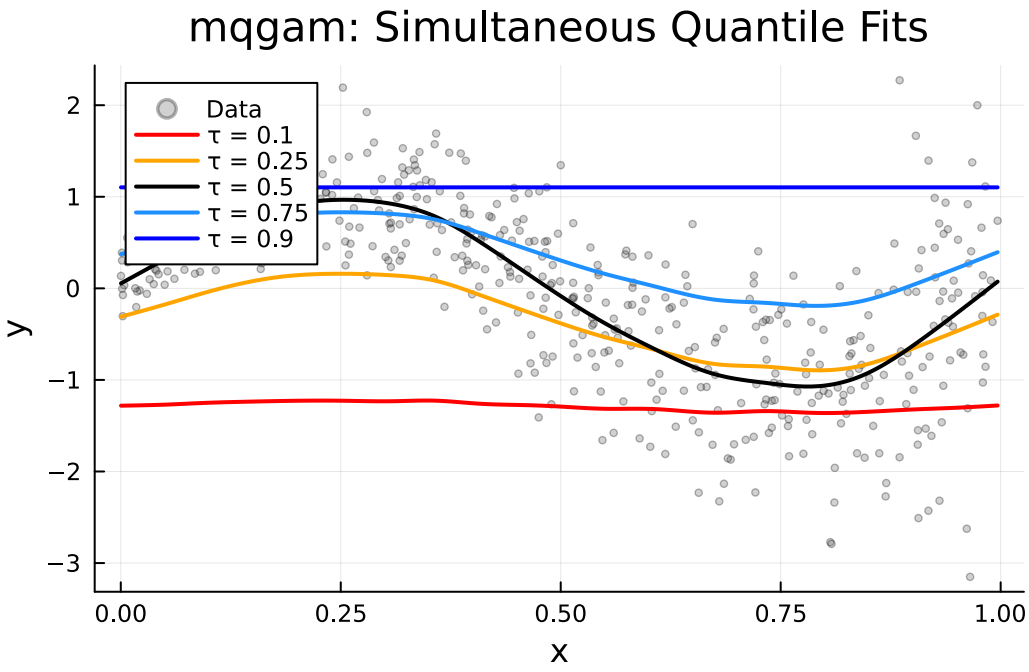
```
Quantiles fitted: [0.1, 0.25, 0.5, 0.75, 0.9]
Number of models: 5
```

Access individual model fits:

```
for qu in quantiles
    yhat = predict(mq.fits[qu])
    println("  = $qu: fitted range [$(round(minimum(yhat), digits=3)), $(round(maximum(yhat),
end
```

```
  = 0.1: fitted range [-1.362, -1.225]
  = 0.25: fitted range [-0.895, 0.159]
  = 0.5: fitted range [-1.072, 0.967]
  = 0.75: fitted range [-0.191, 0.831]
  = 0.9: fitted range [1.103, 1.103]
```

```
qcolors = Dict(0.1 => :red, 0.25 => :orange, 0.5 => :black, 0.75 => :dodgerblue, 0.9 => :blue
scatter(x, y, color=:grey40, markersize=2, alpha=0.3,
    xlabel="x", ylabel="y", title="mqgam: Simultaneous Quantile Fits",
    label="Data", legend=:topleft)
for qu in quantiles
    yhat = predict(mq.fits[qu])[ord]
    plot!(x[ord], yhat, color=qcolors[qu], linewidth=2, label="  = $qu")
end
current()
```



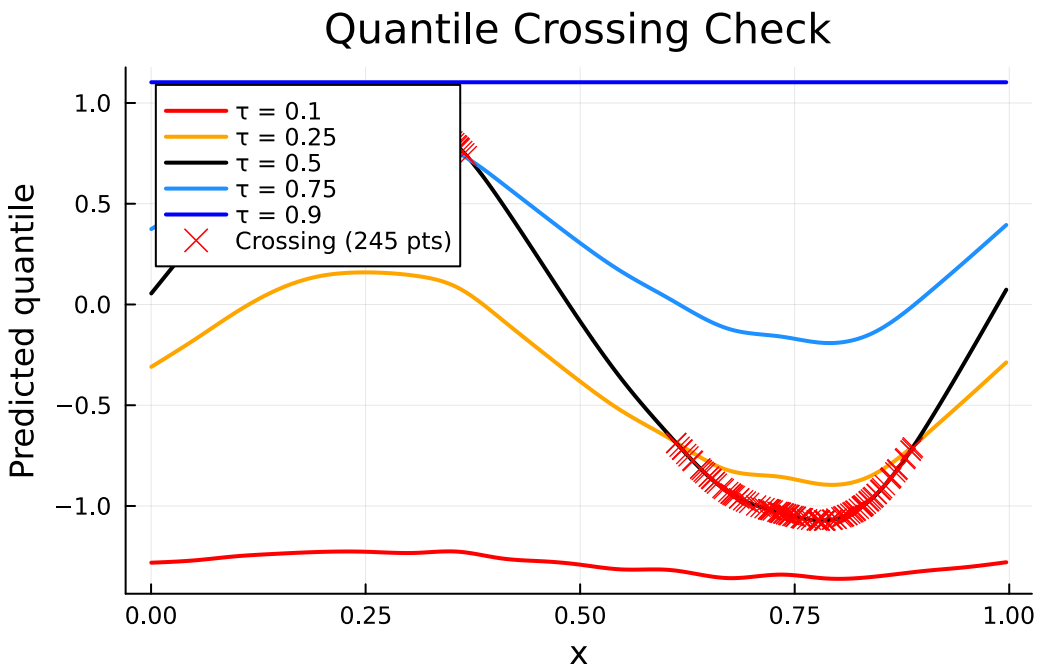
Quantile crossing check

A known issue with independent quantile fits is **quantile crossing**, where estimated quantile curves cross each other. Let's check:

```
x_sorted = x[ord]
predictions = Dict(qu => predict(mq.fits[qu])[ord] for qu in quantiles)

n_crossings = 0
for i in 1:n
    vals = [predictions[qu][i] for qu in quantiles]
    if !issorted(vals)
        n_crossings += 1
    end
end
println("Quantile crossings: $n_crossings out of $n observations ($(round(100*n_crossings/n,
```

```
Quantile crossings: 245 out of 500 observations (49.0%)
```

```
qcolors = Dict(0.1 => :red, 0.25 => :orange, 0.5 => :black, 0.75 => :dodgerblue, 0.9 => :blue
p = plot(xlabel="x", ylabel="Predicted quantile", title="Quantile Crossing Check",
    legend=:topleft)
for qu in quantiles
    plot!(x_sorted, predictions[qu], color=qcolors[qu], linewidth=2, label="  = $qu")
end
crossing_flags = [!issorted([predictions[qu][i] for qu in quantiles]) for i in eachindex(x_so
if any(crossing_flags)
    idx = findall(crossing_flags)
    scatter!(x_sorted[idx], [predictions[0.5][i] for i in idx],
        color=:red, markersize=5, markershape=:xcross,
        label="Crossing ($(length(idx)) pts)")
end
p
```



Verify quantile coverage

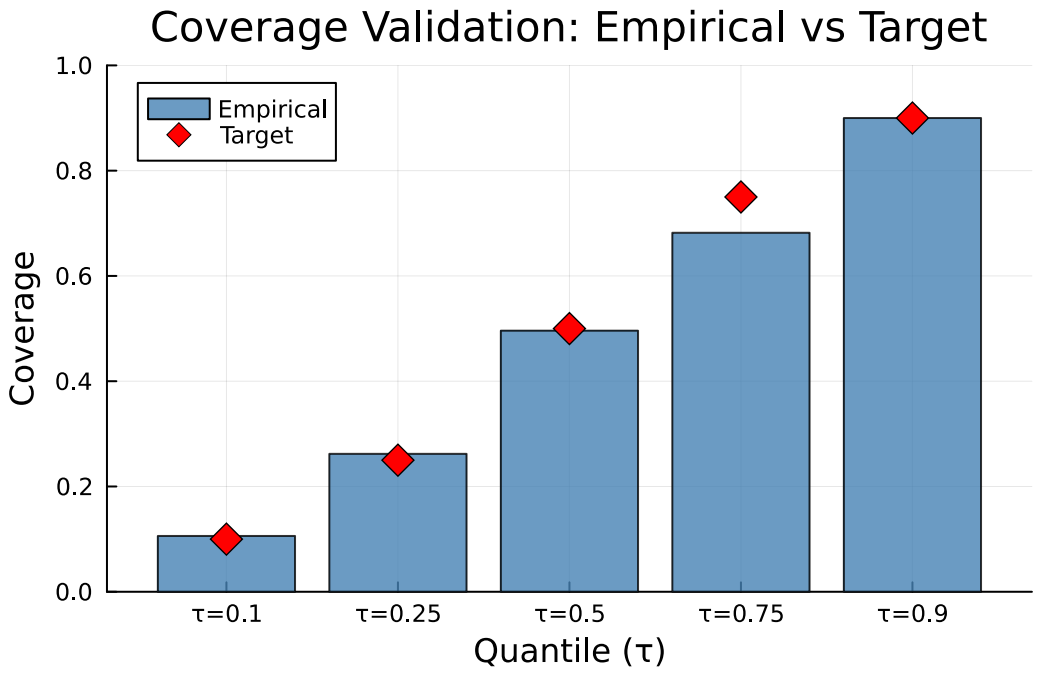
For well-calibrated quantile estimates, approximately $\tau \times 100\%$ of observations should fall below the τ-th quantile curve:

```
for qu in quantiles
    yhat = predict(mq.fits[qu])
    coverage = mean(y .< yhat)
    println("  = $qu: empirical coverage = $(round(coverage, digits=3)) (target = $qu)")
end
```

```
  = 0.1: empirical coverage = 0.106 (target = 0.1)
  = 0.25: empirical coverage = 0.262 (target = 0.25)
  = 0.5: empirical coverage = 0.496 (target = 0.5)
  = 0.75: empirical coverage = 0.682 (target = 0.75)
  = 0.9: empirical coverage = 0.9 (target = 0.9)
```

```
coverages = [mean(y .< predict(mq.fits[qu])) for qu in quantiles]
xs = 1:length(quantiles)
qlabels = [" =$(qu)" for qu in quantiles]
bar(xs, coverages, label="Empirical", color=:steelblue, alpha=0.8,
```

```
    xlabel="Quantile ( )", ylabel="Coverage",
    title="Coverage Validation: Empirical vs Target",
    xticks=(xs, qlabels), ylim=(0, 1), legend=:topleft)
scatter!(xs, collect(quantiles), label="Target", color=:red,
    markersize=8, markershape=:diamond)
```



The ELF family

The Extended Log-F family used internally can be accessed directly:

```
elf = ELFFamily(qu=0.5, co=0.1)
println("ELF family for median: quantile = ", elf.qu)
```

```
ELF family for median: quantile = 0.5
```

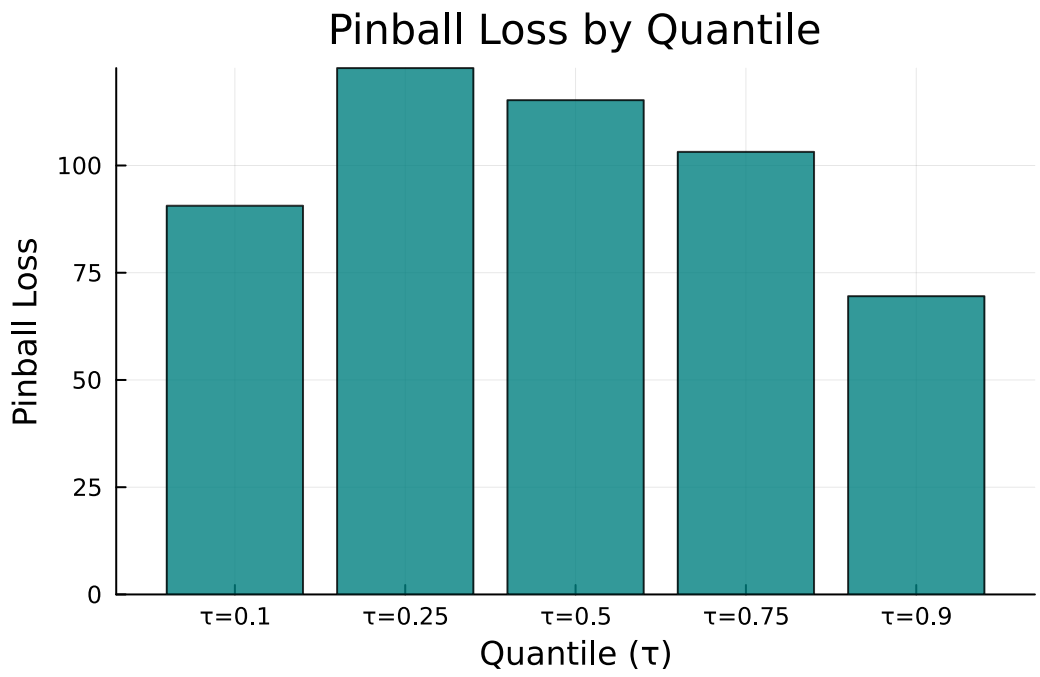
The pinball loss for a given quantile τ:

```
pl = pinball_loss(y, predict(mq.fits[0.5]), 0.5)
println("Pinball loss ( =0.5): ", round(pl, digits=3))
```

```
for qu in quantiles
    pl = pinball_loss(y, predict(mq.fits[qu]), qu)
    println("Pinball loss ( =$qu): $(round(pl, digits=3))")
end
```

```
Pinball loss ( =0.5): 115.234
Pinball loss ( =0.1): 90.617
Pinball loss ( =0.25): 122.715
Pinball loss ( =0.5): 115.234
Pinball loss ( =0.75): 103.175
Pinball loss ( =0.9): 69.533
```

```
losses = [pinball_loss(y, predict(mq.fits[qu]), qu) for qu in quantiles]
xs = 1:length(quantiles)
qlabels = [" =$(qu)" for qu in quantiles]
bar(xs, losses, color=:teal, alpha=0.8, legend=false,
    xlabel="Quantile ( )", ylabel="Pinball Loss",
    title="Pinball Loss by Quantile",
    xticks=(xs, qlabels))
```



Summary

Feature	GAM.jl	R qgam
Single quantile	`qgam(formula, data, )`	`qgam(formula, data=dat, qu= )`
Multiple quantiles	`mqgam(formula, data, [ , , ...])`	`mqgam(formula, data=dat, qu=c(...))`
Loss function	ELF (Extended Log-F)	ELF
Smoothing	Automatic (REML-based)	Automatic
Pinball loss	`pinball_loss(r, )`	`pinLoss(...)`
Formula syntax	`@gam_formula(y ~ s(x, k=20))`	`y ~ s(x, k=20)`

Quantile GAMs extend the GAM framework to model the full conditional distribution, making them especially valuable for heteroscedastic data where the spread changes with covariates.